

Q2 Stemming

What is Stemming?

→ The process of converting a word into its

root form or base form.

→ It removes suffixes like ing, ed Ex Running, Runs, Runner

→ Improves search.

Porter Stemmer.

Stemmed to Run.

→ Applies a series of rules that strip common suffixes.

Text: Dave is a high school student

who likes to study Math. He is currently

preparing for his final year exams".

Original Stemmed.

Dave → Dave

is → is

a → a

high → high

school → school

Student → student

who → who.

likes → like

to → to

study → study

math → math.

• — •

He → He

is → is

currently → currentl;

→ current

preparing → prepar

for → for

his → his

final → final

year → year.

exams → exam.

Q2

Stemming

Stemming is a text normalization process to reduce a word to its ^{root} or base form.

It removes inflectional or derivational endings such as -ing, -ed, -s, ly etc without necessarily producing a dictionary valid word.

- Stemming treats different morphological forms of a word as the same form.

ex connect, connected, connecting, connections
— connect

- Reduces vocabulary size and improves search engine efficiency.

Porter Stemming

- is one of the most widely used stemming algorithms in NLP.

- It applies a sequence of rule based suffix stripping steps to reduce English words to stems.

Main steps of Porter Algorithm.

step 1: Handles plurals and past participles

- Removes -s, es, ed, -ing, -ly

Word	Operation	Stem
cats	remove -s	cat
bones	remove -es	bon
Painted	remove -ed	paint
running	remove -ing	run
quiescely	remove -ly	quiesc

step 2: Converts common suffixes to shorter forms.

Word	Operation	Stem
relational	-ational → ate	relate
ational	-tional → tion	nation

Step 3: Removes or replaces derivational suffixes.

Word	Operation	stem.
electrical	-ical -ic	electr
finalize	Remove	final.
formalize	form -ize	formal.

Step 4: Removes noun and adjective endings.

Word	Ending removed	stem.
obedience	-ance	obedi
existence	-ence	exist
Enjoyment	-ment	enjoy
Socialism	-ism	social.
action	-ion	act

Step 5: Removes final 'e' and double consonants

Word	Rule	stem.
definite	remove e	definit
create	remove e	creat
hopping		hop.

From Passage

Word

Stem

Porter Rule
Applied.

Renowned

Renown

Remove ed

enquire

enquir

Remove e

meticulously

meticul

Remove ly, then ts

Vibrant

Vibrant

No change

ingredients

ingredi

Remove s, then ent

replicating

replic

Remove ing & then at

techniques

techniqu

Remove es.

Q1 Content Free Grammar./ Phrase Structure Grammar/
BNF.

$G = \{ T, N, R, S \}$.

$S \rightarrow$ start symbol.

$T \rightarrow$ Terminal (All English words)

$N \rightarrow$ Nonterminals.

$R \rightarrow$ Set of Rule.

$R =$ predefined Rules.

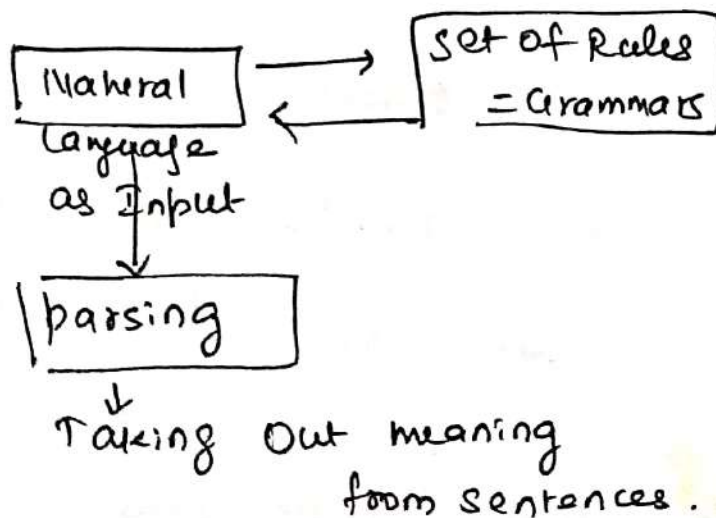
$S \rightarrow NP$		$Np \rightarrow \text{Noun.}$		$vp \rightarrow \text{verb}$
$S \rightarrow NP \quad vp.$		$Np \rightarrow \text{Det Nom}$		$vp \rightarrow \text{verb Np}$
$S \rightarrow vp$		$Nom \rightarrow \text{Noun}$		
$S \rightarrow \text{Aux Np vp.}$		$Nom \rightarrow \text{Noun NOM}$		

$\text{Det} \rightarrow \text{that} / \text{this} / \text{the} / \text{a}$

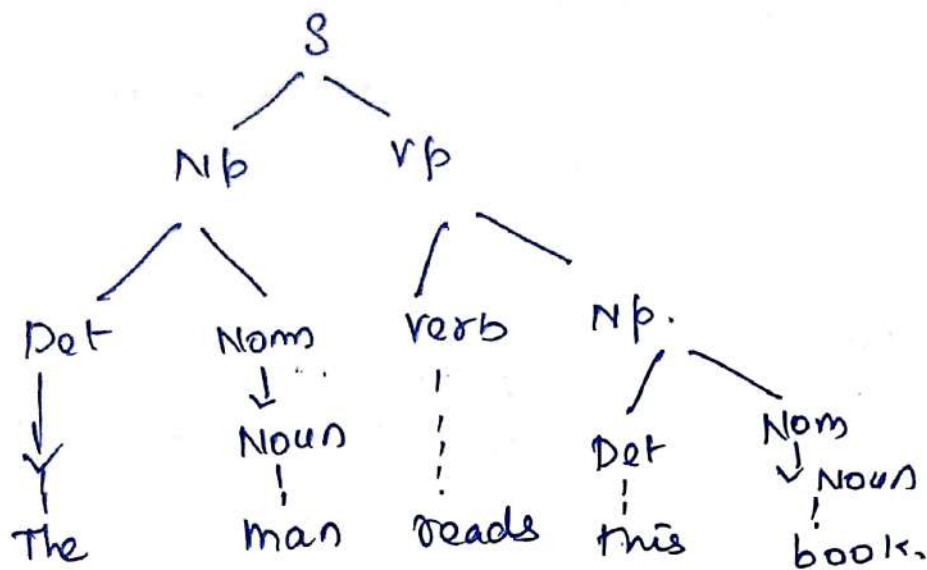
$\text{Noun} \rightarrow \text{books} / \text{Man}$

$\text{Verb} \rightarrow \text{include} / \text{read}$

$\text{Aux} \rightarrow \text{does.}$

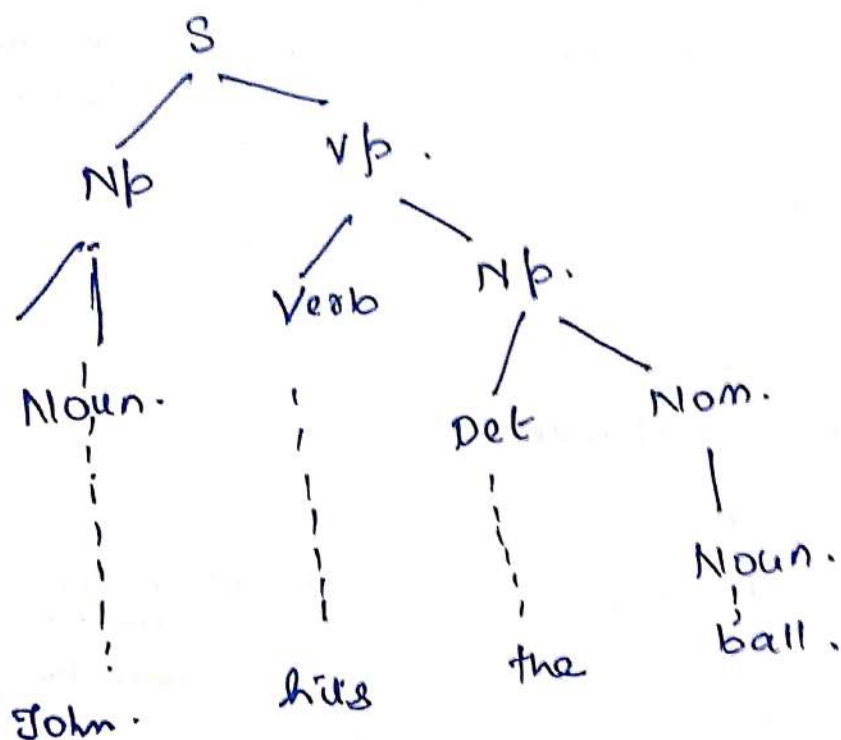


The man reads this book
 Det Noun Verb Det Noun

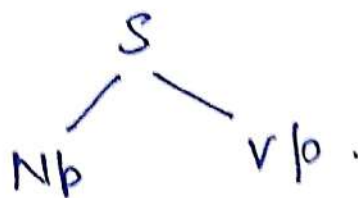


- (1) $S \rightarrow Np \quad vp$
- (2) $\rightarrow Det \quad Nom \quad vp$
- (3) $\rightarrow The \quad Noun \quad vp.$
- (4) $\rightarrow The \quad man \quad vp$
- (5) $\rightarrow The \quad man \quad Verb \quad Np.$
- (6) $\rightarrow The \quad man \quad reads \quad Np.$
- 7 $\rightarrow The \quad man \quad reads \quad Det \quad Nom$
- 8 $\rightarrow The \quad man \quad reads \quad this \quad Noun.$
- 9 $\rightarrow The \quad man \quad reads \quad this \quad Noun$
- 10 $\rightarrow The \quad man \quad reads \quad this \quad book.$

② John hits me ball
Np Verb Det Noun.



(over) ③ He is a good boy
PRP VBZ Det JJ. Noun.



Parsing - A process of analyzing a sentence grammatical structure.

→ Top Down parsing (BFS)

→ (Start with start symbol)

→ Goal directed. → Rewrite the rules according to the goal

→ Bottom up parsing (stack).

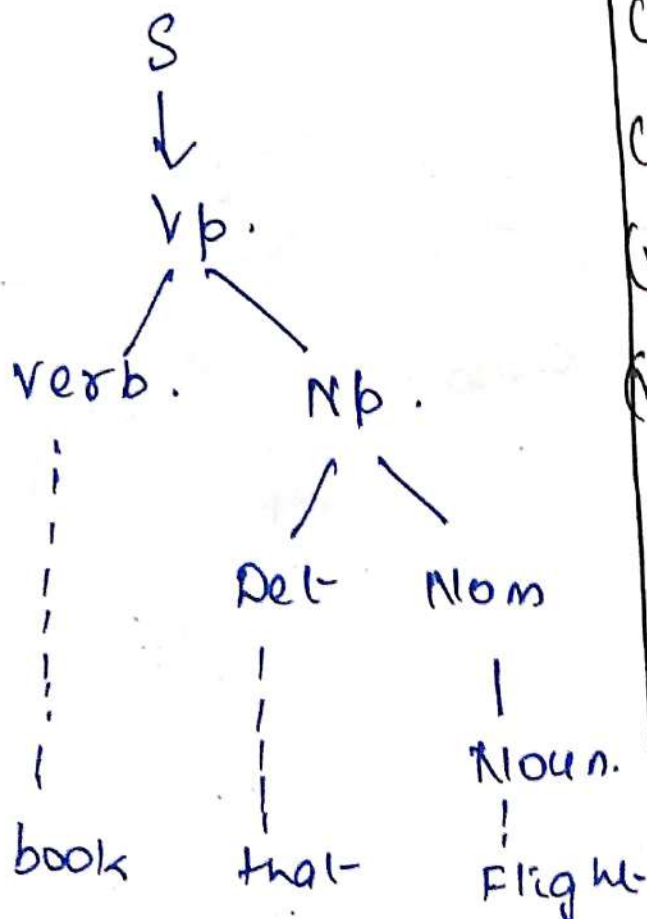
→ Data driven.

(Start with the Terminal!)

String / sentence given.

Q book that flight
verb Det. Noun.

Top Down parsing



Bottom up. (stack)

() book the flight

(book) the flight verb → book

(verb) the flight

(verb the) flight Det → that

(ver Det) flight

(verb Det flight) Noun → flight

(ver Det Noun)

verb Nom

Vp

